VALIDATION SPECIFICATION (v1)

Initial Validation Planning — Requirements Phase

Project: Smart Expense & Budget Management Application

Group Number: G-15

Authors:
Rhea Sushanth – PES1UG24CS371
Sarthki Vachhani – PES1UG24CS422
Sai Manogna BV – PES1UG24CS399

Appendix D: Validation Specification — Initial Planning

At this requirements stage, the validation specification is planning-focused: it explains how functional, security and non-functional requirements will eventually be validated, including planned security testing. This is not the final test report (see the companion Test Plan for that).

Validation vs. Verification: As distinguished in the course: Validation determines whether the requirements, if implemented, will solve the right problem and satisfy the intended user's needs. Verification determines whether the requirements have been specified correctly. Reviews are used for both.

Planned Validation Techniques:

- Requirement reviews: peer walkthroughs within the team, and instructor/evaluator review at each deliverable checkpoint.
- Prototyping: low-fidelity mockups of the transaction-entry and dashboard screens, which involve the most user interaction, to confirm the team and stakeholders share the same interpretation before implementation.
- Model validation: checking that the use-case model (Section 7.1) and the ER/domain model (Section 7.2) are internally consistent and jointly cover every functional requirement in Section 4.
- Acceptance-criteria checks: matching each sprint's user-story acceptance criteria (Appendix E) against the functional requirement(s) it is meant to satisfy.

Validation Plan by Requirement Category:

Requirement Category	**Planned Validation**	**Example**
Functional	Black-box functional tests	Valid/invalid expense entry, budget creation
Security	Authentication/authorization testing + planned penetration testing (e.g., OWASP-style checks for injection, broken access control, and auth bypass)	Customer attempts to access admin category endpoint
Performance	Load/response-time test	Concurrent dashboard/report requests
Usability	Task-based walkthrough	New user records a basic expense
Reliability	Failure/retry scenarios	Re-importing the same CSV file
Unit	Automated unit tests + coverage	Budget spend and alert-threshold calculation